Neural Activity Differences in the Retrieval of Memories With and Without Social Interaction

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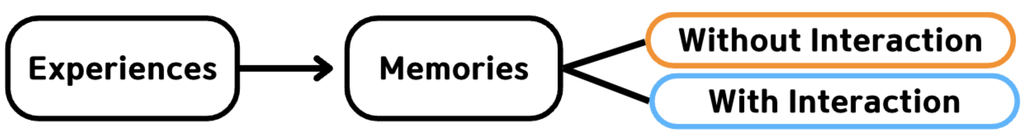
INTRODUCTION

We experience diverse situations daily, from self-reflective introspection to social interactions and tasks.

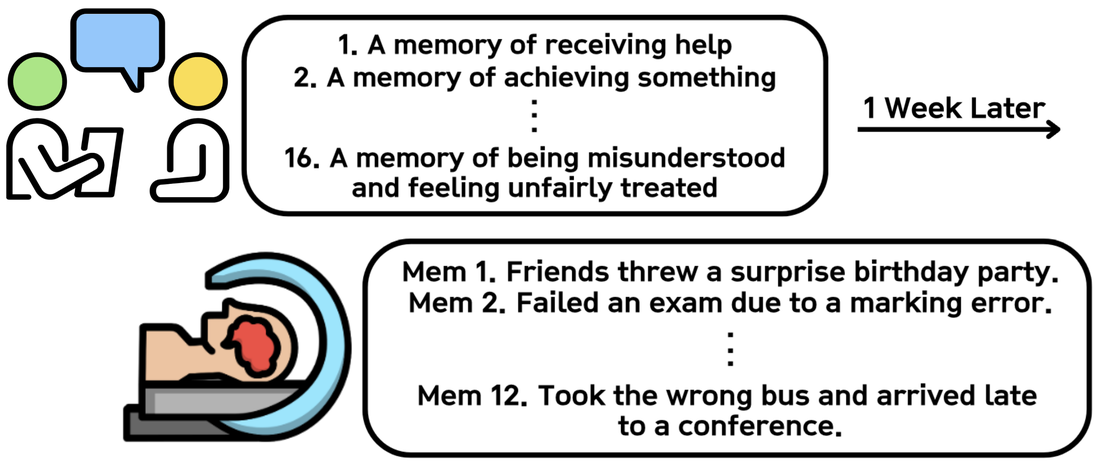
Introspection is known to involve the default mode network (DMN), which consists of key hubs such as the posterior cingulate cortex (PCC), closely connected to the inferior parietal lobule (IPL), and the medial prefrontal cortex (mPFC ; Buckner et al., 2008). In contrast, social activities are associated with regions such as the amygdala, orbitofrontal cortex (OFC), and insula (Adolphs, 2009).

Some of these moments are encoded as memories. Do similar neural mechanisms underlying these situations also play a role during memory retrieval? This study aims to investigate the representational differences in the retrieval of memories with and without social interaction.



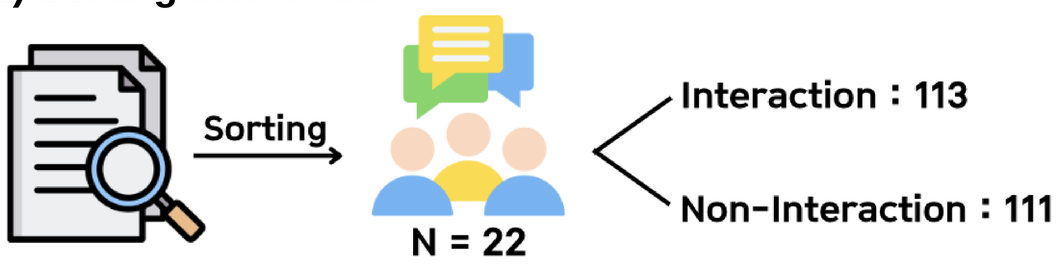
METHODS

1) Interview & fMRI Experiment



Each participant recalled 12 distinct memories, four times each.

2) Sorting Memories



Sorting criterion: Whether there was direct communication with others, such as conversations or message exchanges.

✓ Memories such as being late for an appointment, which involve interaction as a background but are experienced alone, were excluded from the analysis.

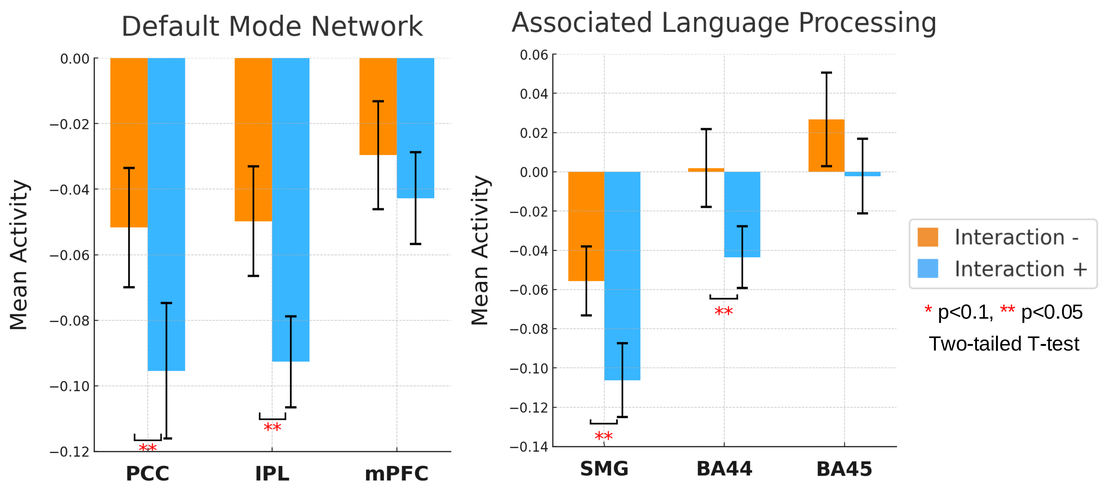
3) fMRI Data Analysis

Mean activation for memories with and without interaction was analyzed for each participant. Furthermore, multivoxel pattern analysis (MVPA) was conducted to examine shared representations of social interaction.

* The participant interview data and fMRI data were provided by Professor Sue-Hyun Lee's "Memory and Cognition Lab."

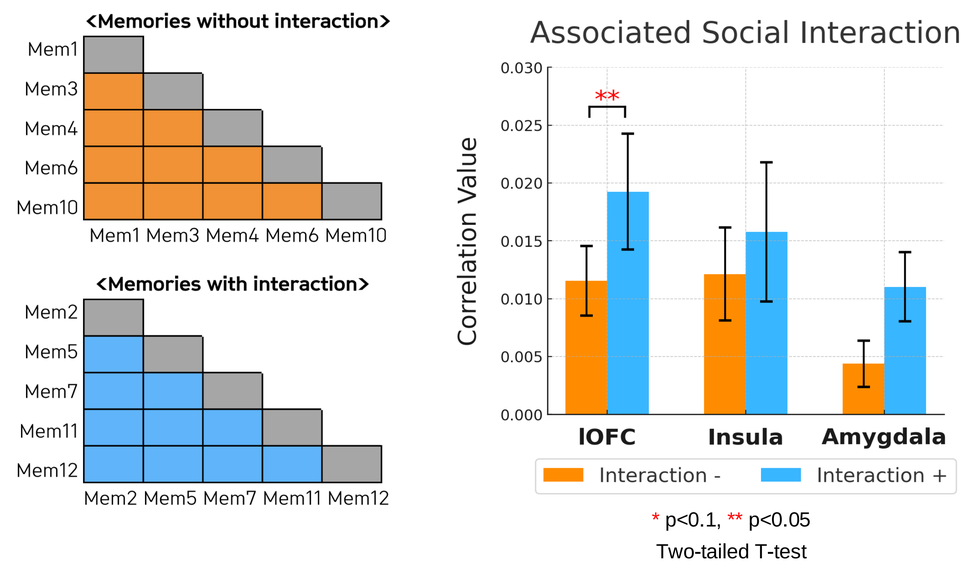
RESULTS

1) Comparison of Neural Activity



Neural Activity data for posterior cingulate cortex (PCC), inferior parietal lobule (IPL), medial prefrontal cortex (mPFC), supramarginal gyrus (SMG), and BA44/BA45 of Broca's area

2) Comparison of Correlations



Correlation data for lateral orbitofrontal cortex (lOFC), insula, and amygdala

DISCUSSIONS

1) Interpretation of Results

- The results highlight the involvement of the **PCC** and **IPL** in self-referential processing and suggest that the **lOFC** plays a critical role in social situational processing.
- Notably, regions associated with language processing showed greater activation during the retrieval of memories without interaction, which is likely attributed to. **self-referential inner speech during the recall of self-focused memories**
- The lack of significant differences in the insula and amygdala likely reflects the absence of empathizing with others in most memories and **the lack of emotional facial expression recognition** during memory retrieval.

2) Future Direction

- Segmenting memories into those that elicit positive emotions and those that evoke negative emotions reduces variability in the analysis and enhances interpretability.

REFERENCES

Adolphs, R. (2009). The social brain: Neural basis of social knowledge. *Annual Review of Psychology, 60*, 693–716.

Buckner, R. L., Andrews-Hanna, J. R., & Schacter, D. L. (2008). The brain's default network: Anatomy, function, and relevance to disease. *Annals of the New York Academy of Sciences, 1124*(1), 1–38.